Shweta Prasad

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Education

- **IMPRS MMFD, University of Tübingen** — Tübingen, Germany
  M.Sc in Neural and Behavioural Sciences; CGPA 1.68 — *Oct. 2022 – Oct. 2024*
- **Ashoka University** — Sonipat, India
  PG Diploma in Advanced Studies and Research in Computer Science; CGPA 3.44/4.0 — *Aug. 2021 – May 2022*
- **Ashoka University** — Sonipat, India
  B.Sc in Computer Science with a Minor in Biology; CGPA: 3.58/4.0 — *Aug. 2018 – May 2021*

Awards and Scholarships

- Best Student Presentation, Lab Rotation Seminar, GTC of Neuroscience, University of Tübingen
- Fully funded M.Sc in Neural and Behavioural Sciences, MPI for Biological Cybernetics
- Full scholarship (Tuition and Residence) Ashoka University

Tools and Frameworks

- **Programming Languages:** Python, MATLAB, C, C++, R, Apps Script
- **Frameworks:** PyTorch, TensorFlow, sklearn for machine learning; C#/.NET for Windows app development
- **Other Tools**: GitHub, Selenium, Excel/G-Sheets, LaTeX, GIMP, CORTEX, Inkscape

Research Experience

- **Waveform feature analysis, Hafed Lab, Center for Integrative Neuroscience** — Tübingen, Germany
  Research Assistant, Supervised by Prof. Dr. Ziad Hafed — *Jul. 2024 –*
  - Identification of functionally different neuron types from action potential waveforms in the rhesus macaque superior colliculus (SC) and primary visual cortex (V1)
  - Automated data curation pipeline & collated a large dataset of SC and V1 neurons
- **Decision making in crows, Nieder Lab, Inst. Neurobiology** — Tübingen, Germany
  Lab-rotation & Master Thesis Student, Supervised by Prof. Dr. Andreas Nieder — *Nov. 2023 – Oct. 2024*
  - Master thesis: Waiting time as a behavioural correlate for decision confidence in a carrion crow (Grade: 1.3)
  - Lab rotation: Behavioural aspects of perceptual decision making in carrion crows
- **Serotonin in patience, Dayan Lab, MPI for Biological Cybernetics** — Tübingen, Germany
  Lab-rotation Student, Supervised by Dr. Kevin Lloyd — *Sep. 2023 – Nov. 2023*
  - Evaluated an average reward RL model of how Serotonin may be modulating patience in interval timing tasks
- **Single Molecule Biophysics, Haldar Lab, Ashoka University** — Sonipat, India
  Software Development Intern, Supervised by Dr. Debayan Gupta & Dr. Shubhasis Haldar — *Dec. 2019 – May 2022*
  - Building and improving covalent magnetic tweezers and associated software(s) for week-long single molecule force-clamp experiments.
  - Learnt protein extraction and purification methods over Summer, 2020
- **Neuroethology Lab, Ashoka University** — Sonipat, India
  Undergraduate Research Assistant to Dr. Bittu K Rajaraman — *Sep. 2018 – May 2022*
  - Studied call pattern generation and song production in *Orthopterans* using a mixture of simulations, electrophysiological and behavioral exploration, to develop circuit models
  - Gained experience in field work and animal maintenance

Publications

- (In preparation) **Prasad, S.**, Malevich, T., Hafed., Z. M. (2026) Predicting cell-type specific function from extracellular spike waveform shapes in the rhesus macaque primary visual cortex and superior colliculus.
- Trottenberg, C., Yu, Y., Zhang, T., Baumann, M. P., Malevich, T., **Prasad, S.**, Hafed, Z. M. (2025) Much higher covariation with foveation timing by superior colliculus than primary visual cortical neuronal activity. *iScience*, DOI: 10.1016/j.isci.2026.115432
- **Prasad, S.**, Jhawar, M. P., Banerjee, S. (2022) Combining Asymmetrical Numerical Systems based codecs and deep probabilistic models for efficient, near-lossless compression of videos. Bachelor thesis, Ashoka University, India.

Teaching

- **Graduate Teaching Assistant**
  University of Tübingen
  - **Neurobiology Practical (Winter 2023)** , Dr. Stephanie Westendorff; Class size: 25; Tasks: Assisting 3rd year students of B.Sc Neurobiology conduct electrophysiological exploration of ground cricket auditory responses
  - **Sensory Systems - I: The Auditory and Vestibular Systems (Winter 2023)** , Class Size: 30; Tasks: holding bi-weekly tutorials for masters students of the Graduate Training Center of Neuroscience
- **Undergraduate Teaching Assistant**
  Dept. of Computer Science, Ashoka Universtiy
  - **Theory of Computation (Spring 2022)** , Dr. Soumyottam Chatterjee; Class size: 25; Student Feedback : 4.5/5; Tasks: holding weekly office hours, setting and grading all assignments
  - **Introduction to Machine Learning (Monsoon 2021)** , Dr. Subhashish Banerjee; Class size: 68; Student Feedback: 4.43/5; Tasks: holding weekly office hours, setting and grading all assignments, facilitating data collection for Ashoka's Faces Dataset
  - **Algorithms Design and Analysis (Spring 2020)** , Dr. Subhash Bhalla; Class Size: 70; Student Feedback: 4.46/5; Tasks: holding weekly office hours, setting and grading all assignments
- **Teaching Assistant**
  Summer School, Neuromatch Academy (Online)
  - **Deep Learning Course (Summer 2021)** , Content by various professors from around the globe; Led 7-14 international UGs and Ph.Ds selected to participate in the *Interactive Track* of the programme; Led daily discussion sessions, taught deep learning tools from ground up on PyTorch, provided project support

Leadership and Extra-Curriculars

- **Student Representative, IMPRS MMFD**
  University of Tuebingen *Mar. 2023 – Mar. 2024*
- **Mentor, Ashoka University Women in STEM**
  Independent, Alumni-run *Oct. 2022 –*
  - Mentoring women and non-binary folk interested in pursuing research careers in STEM
- **Advisor, Women in Computing Society**
  Ashoka Universtiy *Sep. 2021 – Jan. 2022*
  - Head, *WiCS Workshop Weekends*; Member since 2018
  - Led 2 *WiCS Annual Cryptic Hunts*, our flagship event
- **Student Representative, Dept. of Computer Science**
  Academic Advisory Board, Ashoka Universtiy *Sep. 2020 – May 2021*
  - Selected by the Head of Department, Dept. of Computer Science
  - Served as primary coordinator for all student–department communication

Workshops, Talks, Conferences, Symposia

- **European Conference on Eye Movements (2026)** Poster – *Classifying extracellular action potential waveform shapes in the rhesus macaque superior colliculus and primary visual cortex*
- **Primate Neurobiology Meeting (2026)** Poster – *Classifying extracellular action potential waveform shapes in the rhesus macaque superior colliculus and primary visual cortex*
- **ZüMüTü Oculomotor and Vestibular meeting (2026)** Talk – *Classifying extracellular action potential waveform shapes in the rhesus macaque superior colliculus and primary visual cortex*
- **Gordon Research Conference on Eye Movements (2025)** Poster – *Classifying extracellular action potential waveform shapes in the rhesus macaque superior colliculus and primary visual cortex*
- **Animal Tracking Workshop (2025)** Organizer, workshop held on 13 & 14 Feb., 2025; Collaboration between the IMPRS MMFD and the *Center for the Advanced Study of Collective Behaviour* (CASCB)
- **Rewilding Ethology (2024)** Workshop at the Max Planck Institute for Animal Behaviour, Konstanz, Germany (fully funded to attend by the CASCB)
- **Systems Neuroscience Symposium (2023, 2024)** Symposia at UKT Tübingen, Germany
- **No Garland Neuroscience (2020)** Conference at IISER Pune, India

Other Work Experience

- **Alumni Relations Office, Ashoka University** — Sonipat, India
  Website & Database Administrator — *Mar. 2021 – Jan. 2022*
  - Manager of alumni databases and the alumni web portal, with 1500+ active users
  - Supervised interns over summer 2021 to clean data and expand existing database by scraping 980+ LinkedIn profiles to collect and organize data on all existing alumni into the most comprehensive database in university history
- **The Neuroscience Outreach Network** — Princeton, NJ, USA
  Community Outreach & Content Coordinator — *Aug. 2020 – Jan. 2021*
  - Project aimed at enabling access to education in neuroscience to students in underserved communities around the world.
  - Tailored virtual/classroom lessons and material for individual grade levels
- **DiverseNeuro.org** — Multiple locations, India
  Student Researcher — *Jun. 2020*
  - A collaborative research venture between IISER Pune and Ashoka University
  - Study on the international academic demographic in neuroscience, aimed at informing inclusive policy-making in academia